Midterm Exam 2: Help Sheet

ECON 441: Introduction to Mathematical Economics — Instructor: Div Bhagia

Limit Definition of the Derivative

$$\frac{dy}{dx} = f'(x_0) = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

Limit of a Function

The limit of a function $f(x)$ exists at $x = a$ if both the right-side and left-side limits at a are equal.

Continuity of a Function

A function $y = f(x)$ is continuous at a if the limit of $f(x)$ at a exists and is equal to the value of the function at a, i.e. $\lim_{x \to a} f(x) = f(a)$.

Product Rule

$$\frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + f'(x)g(x)$$

Quotient Rule

$$\frac{d}{dx}\frac{f(x)}{g(x)} = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$$

Elasticity

$$\varepsilon = \frac{\text{Percentage change in } y}{\text{Percentage change in } x} = \frac{dy/y}{dx/x} = \frac{dy}{dx} \cdot \frac{x}{y} = \frac{d \ln y}{d \ln x}$$

Integration

Integration reverses differentiation: if $F'(x) = f(x)$, then

$$\int f(x)\,dx = F(x) + c \qquad \int_a^b f(x)\,dx = F(b) - F(a)$$

Partial Derivative

If x_1 changes by Δx_1 but all other variables remain constant:

$$\frac{\Delta y}{\Delta x_1} = \frac{f(x_1 + \Delta x_1, x_2, \ldots, x_n) - f(x_1, x_2, \ldots, x_n)}{\Delta x_1}$$

The partial derivative of y with respect to x_i:

$$\frac{\partial y}{\partial x_i} = f_i = \lim_{\Delta x_i \to 0} \frac{\Delta y}{\Delta x_i}$$

Gradient Vector

$$\nabla f(x_1, x_2, \ldots, x_n) = [f_1, f_2, \ldots, f_n]^T$$

Total Differential

$$df = f_1\,dx_1 + f_2\,dx_2 + \cdots + f_n\,dx_n = \sum_{i=1}^{n} f_i\,dx_i$$

Total Derivative

Total derivative with respect to x_1:

$$\frac{df}{dx_1} = f_1 + f_2\frac{dx_2}{dx_1} + \cdots + f_n\frac{dx_n}{dx_1}$$

If x_i does not depend on x_1 then $dx_i/dx_1 = 0$.

If f does not directly depend on x_1 then $f_1 = 0$.